

UMC743: Quantitative and Statistical Methods for Finance

L	T	P	Cr
2	0	2	3

Course Objective: Understanding quantitative and statistical methods used for finance and derivatives.

Syllabus

Refresher on Statistics: Correlation, OLS regression, probability distributions and moments, using Microsoft Excel for statistical calculations and interpretations

Option Pricing Introduction to option pricing models, formulae and derivation, option Greeks, risk management using options

Financial Time Series: Introduction to time series, types, univariate and multivariate time series models, autocorrelation, AR models, MA models, ARMA models, ARIMA models, stationary series, unit-root

Volatility: Meaning of volatility, types, methods of calculation, volatility models, estimation

Laboratory Work: To use the proprietary software in live international derivatives

1. Introduction to other derivative products
2. Refining trading strategies created in previous courses.
3. Practicing the strategies on the proprietary software.
4. Application of the quantitative and statistical methods
5. Introduction to algorithmic trading on the proprietary software

Recommended Prerequisites: Courses – Finance, Accounting and Valuation + Financial Markets and Portfolio Theory + Derivatives Pricing, Trading and Strategies

Course Learning Objectives (CLO)

After the completion of the course, the student will be able to:

1. Explain and apply the time series models learnt for financial data
2. Explicate and apply PCA for financial data
3. Explain and apply option pricing models to calculate prices of options
4. Explicate types of volatility and volatility models
5. Apply the basic methods learnt to estimate volatility

Reference Books:

1. Introduction to Time Series Analysis and Forecasting - Douglas C. Montgomery, Cheryl L. Jennings
2. Fundamentals of Futures and Options Markets - John C. Hull